

29

MONDAY

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الاثنين

A.C Fundamentals

Circuit A.C is current that periodically reverses direction. The sinusoidal alternating voltage can be expressed by the equation

$$e = E_m \sin \omega t$$

e = Instantaneous value of alternating voltage

E_m = maximum voltage

ω = angular velocity of the coil

Also Sinusoidal current can be expressed in same way as voltage

$$i = I_m \sin \omega t$$

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الأحد

Remember that;

$$e = E_{\max} \sin \omega t$$

$$\omega = 2\pi f$$

$$f = \frac{P}{2} \left(\frac{N}{60} \right) \rightarrow \text{r.p.m}$$

$$E_{\max} = NAB\omega$$

N : no. of turns

A : cross sectional area (m^2)

B : flux density (wb/m^2)

ω = angular velocity ($\text{rad}/\text{sec.}$)

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أيلول September 9

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The effective value or RMS value.

The effective value or r.m.s value of an alternating current is the value of the direct current which produces in the same conductor the same amount of heat in the same time

$$I_{\text{eff}} = I_{\text{r.m.s}} = \frac{I_m}{\sqrt{2}}$$

$$= 0.707 I_m$$

I_{av} : average value of a.c

$$I_{\text{av}} = \frac{2}{\pi} I_m = 0.637 I_{\text{max}}$$

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الخميس

Form Factor:

The ratio of r.m.s. value to the average value

value of an alternating current gives an indication of it's wave shape and is known as its form factor and denoted by k .

$$K = \frac{I_{rms}}{I_{av}} = \frac{\frac{I_m}{\sqrt{2}}}{\frac{2}{\pi} I_m} = \frac{0.707 I_m}{0.637 I_m} = 1.11$$

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الجمعة

$$K = 1.11$$

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Wed.		24	17	10	الأربعاء
Thu.		25	18	11	الخميس
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الأربعاء

Peak Factor

It is defined as the ratio of maximum value to rms value

$$\text{Peak Factor} = \frac{I_m}{I_{rms}} = \frac{I_m}{\frac{I_m}{\sqrt{2}}} = \sqrt{2} = 1.414$$

Example (1):

Calculate the maximum value of emf generated in a coil which is rotating at 50 r.p.s in a uniform magnetic field of 0.8 wb/m^2 . The coil is wound on a square former having 5cm length wound with 300 turns.

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الثلاثاء

given s

$$f = 50 \text{ r.p.s} \quad B = 0.8 \text{ wb/m}^2$$

$$N = 300 \text{ turns} \quad A = 5 \text{ cm} \times 5 \text{ cm}$$

$$E = ? \quad = 25 \text{ cm}^2$$

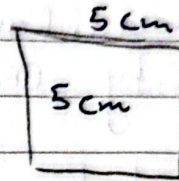
$$= 25 \times 10^{-4} \text{ m}^2$$

$$E_m = NBA \omega$$

$$E_m = NBA 2\pi f$$

$$= 300 \times 0.8 \times 25 \times 10^{-4} \times 2\pi \times 50$$

$$= 188.49 \text{ volt}$$



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أيلول September 9

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Wed.		24	17	10	الأربعاء
Thu.		25	18	11	الخميس
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Example (2)

A rectangular coil measuring 30cm by 20cm and having 40 turns is rotated about an axis coinciding with one of its longer sides at a speed of 1500 r.p.m in an uniform magnetic field of flux density 0.075 Wb/m^2

calculate :

(i) maximum value

(ii) the value of emf when the coil makes an angle of 45° with the field

(iii) frequency of the emf

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الأحد

givens :-

$$A = 30 \text{ cm} \times 20 \text{ cm} = 600 \text{ cm}^2 = 600 \times 10^{-4} \text{ m}^2,$$

$$P = 1500 \text{ r.p.m} \quad B = 0.075 \text{ wb/m}^2$$

$$N = 40 \text{ turns}$$

$$(iii) f = \frac{P}{2} \cdot \left(\frac{N}{60} \right) = \frac{2}{2} \left(\frac{1500}{60} \right) = 25 \text{ Hz}$$

$$(i) E_m = NBA 2\pi f = 40 \times 0.075 \times 600 \times 10^{-4} \times 2\pi \times 25$$

$$E_m = 28.27 \text{ V}$$

$$(ii) e = E_m \sin \omega t$$

$$e = 28.27 \sin 45$$

$$e = 19.98 \text{ volt}$$

الأسبوع ٢٩

September 9 أيلول

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Wed.	24	17	10	03	الأربعاء
Thu.	25	18	11	04	الخميس
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السبت

Example(3)

Find the instantaneous value of

$$v(t) = 5 \sin(2\pi \times 10^4 t)$$
when ① $t = 0 \text{ sec}$ ② $t = 25 \mu\text{s}$

$$v(t) = 5 \sin(2\pi \times 10^4 t)$$

① at $t = 0 \text{ sec}$

$$v(t) = 5 \sin(2\pi \times 10^4 \times 0)$$

$$v(t) = 5 \sin 0 = 0$$

② at $t = 25 \mu\text{s}$

$$v(t) = 5 \sin(2\pi \times 10^4 \times t)$$

$$= 5 \sin(2\pi \times 10^4 \times 25 \times 10^{-6})$$

$$= 5 \sin(0.5\pi) \rightarrow 0.5\pi \times \frac{180}{\pi}$$

$$= 90^\circ$$

$$= 5 \sin(90^\circ)$$

$$= 5 \text{ volt}$$

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الخميس

Example (4) :

The equation of an alternating current
 $i = 42.42 \sin 628t$

Determine :

(a) it's maximum value

(b) it's frequency

(c) it's rms value

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الجمعة

(d) it's average value

(f) it's peak factor

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September 9 أيلول

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الأربعاء

Solution

$$i = 42.42 \sin 628t$$

$$i = I_m \sin \omega t$$

$$(a) \quad I_m = 42.42 A$$

$$(b) \quad \omega = 628$$

$$2\pi f = 628$$

$$f = \frac{628}{2\pi} = 99.94 \text{ Hz}$$

$$(c) \quad I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707 I_m = 0.707 \times 42.42$$

$$= 29.99 A$$

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الثلاثاء

Follow

$$(d) I_{av} = \frac{2}{\pi} I_m = \frac{2}{\pi} \times 42.42 = 27.005A$$

$$(e) k = \frac{I_{rms}}{I_{av}} = \frac{29.99}{27.005} = 1.11$$

$$(f) \text{ Peak Factor} = \frac{I_m}{I_{rms}}$$

$$= \frac{I_m}{\frac{I_m}{\sqrt{2}}} = \sqrt{2} = 1.414$$

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الإثنين

Example (5):

Write the mathematical expression
for a 25 Hz sinusoidal voltage
having peak value 80 V

$$V = V_m \sin \omega t$$

$$V_p = V_m = 80 \text{ V}$$

$$\omega = 2\pi f = 2\pi \times 25 = 50\pi$$

$$V = 80 \sin 50\pi t$$

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الأحد

Exercise

An alternating current of a frequency, 60 Hz has a maximum value of 120 A

(i) write down the equation for the instantaneous value

(ii) find the instantaneous value after $\frac{1}{360}$ seconds

(iii) Time taken to reach 96 A for the first time

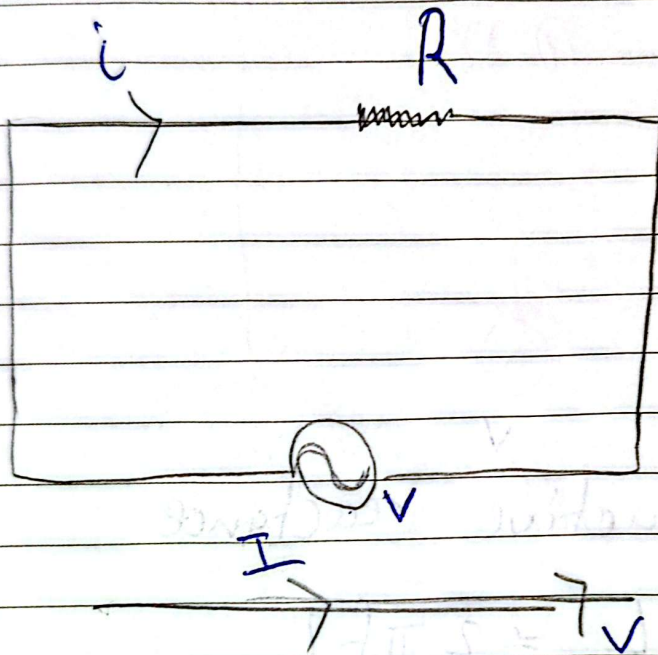
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الخميس

ohmic - circuit



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الجمعة

$$i = \frac{V}{R} \quad V = iR$$

V and I are in phase ($\theta = 0$)

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أيلول September 9

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ملاحظات

Sat.	27	20	13	06	السبت
Sun.	28	21	14	07	الأحد
Mon.	29	22	--		

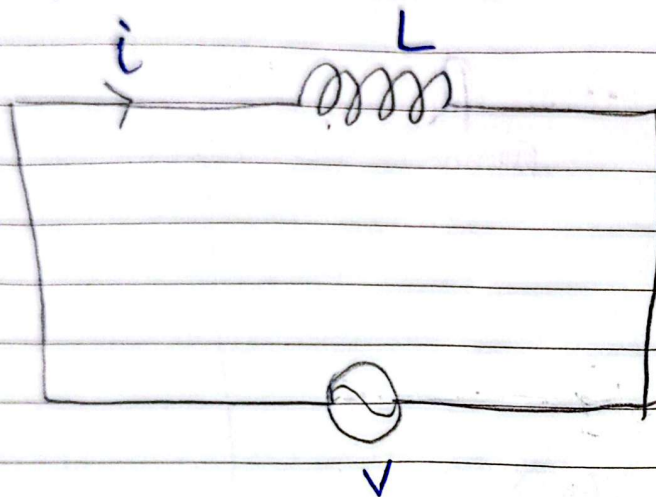
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الأربعاء

Inductive circuit

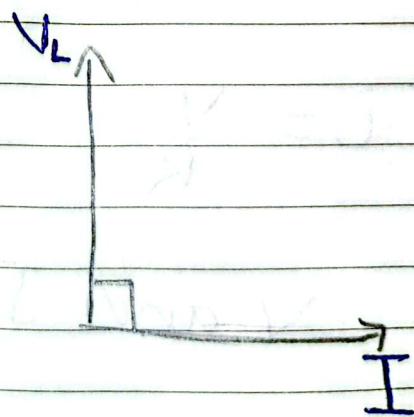
 $X_L = \text{Inductive reactance}$

$$X_L = \omega L = 2\pi f L$$

$$\hat{I} = \frac{V}{X_L} \Rightarrow V = \hat{I} \cdot X_L$$

Voltage lead current
by 90° or current

lag the voltage by $\frac{\pi}{2}$



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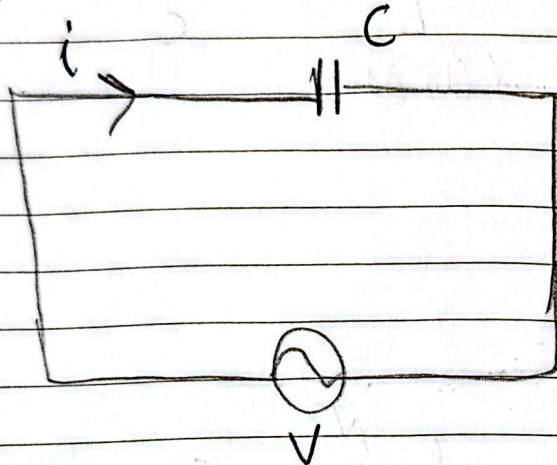
NOTES

ملاحظات

تشرين أول October 10

Sat.	25	18	11	04	الجمعة
Sun.	26	19	12	05	الأحد
Mon.	27	20	13	06	الاثنين
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Capacitive circuit

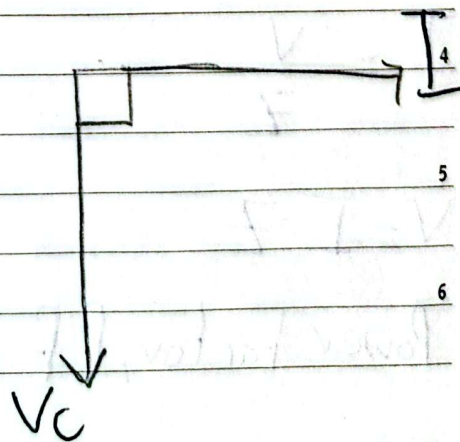


$X_c = \text{capacitive reactance}$

$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$I = \frac{V}{X_c} \Rightarrow V = i X_c$$

Current lead voltage
by 90° or voltage lag
the current by 90°



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أيلول September 9

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Tue.	30	23	16	09 02	الثلاثاء
Wed.		24	17	10 03	الأربعاء

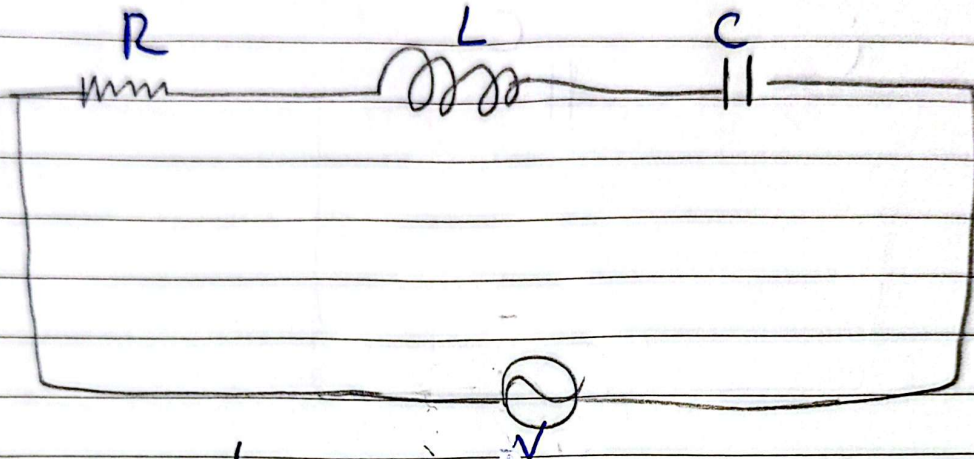
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الاثنين

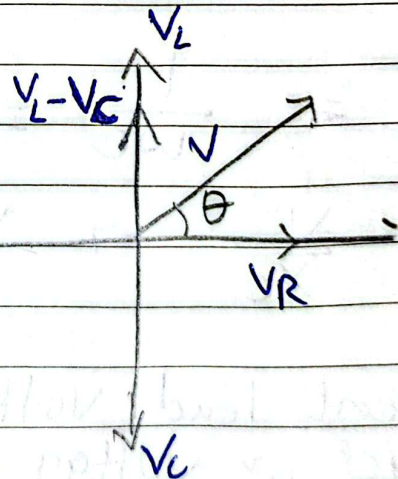
R-L-C - series circuit



Z is impedance of the circuit
(Total resistance)

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$



$$I = \frac{V}{Z}$$

$$V = I Z$$

$$\text{Power Factor, P.F.} = \cos \theta = \frac{R}{Z}$$

Week 3

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Wed.	29	22	15	08	الأربعاء
				01	الخميس

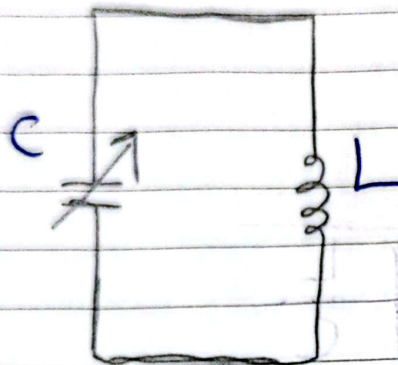
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الأحد

Tunning circuit



At resonance

$$X_L = X_C$$

$$2\pi fL = \frac{1}{2\pi fC}$$

$$4\pi^2 f^2 LC = 1$$

$$f^2 = \frac{1}{4\pi^2 LC} \Rightarrow f = \frac{1}{2\pi \sqrt{LC}}$$

f is called resonance frequency
at resonance $X_L = X_C$ then $Z = R$
It will be maximum at resonance

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September 9 أيلول

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السبت

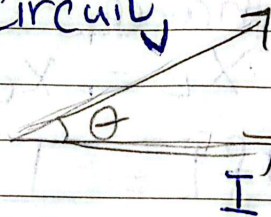
Resonance frequency

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$Q\text{-Factor} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Power in A.C. circuit,

$$P = VI \cos \theta$$



$$\text{efficiency } \eta = \frac{P_{out}}{P_{in}}$$

h.p. = horse Power = 746 watt

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الخميس

Example (1):

A series circuit contains a resistance of 4Ω and inductance of 0.5 H and a variable Capacitance across 100 V , 50 Hz supply. Find

(a) The capacitance for getting resonance

(b) The P.d across inductance and Capacitance

(c) The Q-Factor

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الجمعة

$$R = 4\Omega$$

$$L = 0.5\text{ H}$$

$$V = 100\text{ V}$$

$$f = 50\text{ Hz}$$

$$\textcircled{1} C = ?$$

$$\textcircled{2} \text{ P.d } V_L < V_C = ?$$

$$\textcircled{3} Q\text{-factor} = ?$$

الأسبوع ٣٦

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ملاحظات

(i) At resonance

$$X_L = X_C$$

$$2\pi fL = X_C$$

$$2\pi \times 50 \times 0.5 = X_C$$

$$X_C = 157.07 \Omega$$

$$X_C = \frac{1}{2\pi fC}$$

$$\frac{1}{2\pi fC} = 157.07$$

$$C = \frac{1}{2\pi f \times 157.07} = \frac{1}{2\pi \times 50 \times 157.07}$$

$$C = 2.026 \times 10^{-5} \text{ F}$$

$$C = 20.26 \times 10^{-6} \text{ F} = 20.26 \mu\text{F}$$

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الثلاثاء

$$(c) V_L = V_C = I X_L = I X_C$$

At resonance

$$Z = R$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

and the current will be maximum

$$I = \frac{V}{R} = \frac{100}{4} = 25 \text{ A}$$

$$P.d = V_L = V_C = I X_L = 25 \times 157.07$$

$$P.d = 3926.75 \text{ V}$$

$$(ii) Q\text{-Factor} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$= \frac{1}{4} \sqrt{\frac{0.5}{20.26 \times 10^{-6}}} = 39.274$$

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September 9 أيلول

NOTES

ملاحظات

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Example (2):

A 200V, 50 Hz supply inductive circuit takes a current of 10A lagging 30°

Find (i) resistance

(ii) reactance

(iii) inductance

Solution

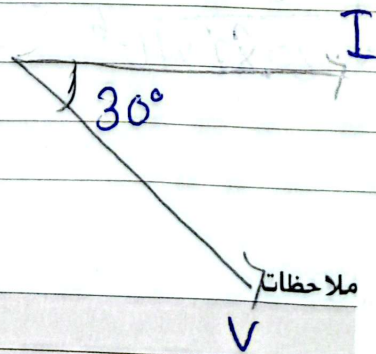
$$V = 200 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$I = 10 \text{ A}$$

$$\theta = 30^\circ$$

(i) $R = ?$ (ii) $X_L = ?$ (iii) $L = ?$



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Week 3

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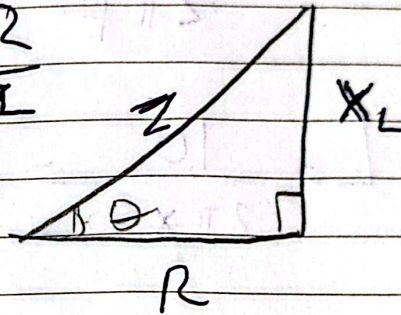
الأحد

follow

$$Z = \frac{V}{I} = \frac{200}{10} = 20 \Omega$$

$$\sin \theta = \frac{X_L}{Z} \quad \cos \theta = \frac{R}{Z}$$

$$X_L = Z \sin \theta$$



$$(i) R = Z \cos \theta$$

$$R = 20 \cos 30^\circ = 20 \times \frac{\sqrt{3}}{2}$$

$$R = 17.32 \Omega$$

$$(ii) X_L = Z \sin \theta$$

$$X_L = 20 \sin 30^\circ = 20 \times \frac{1}{2}$$

$$X_L = 10 \Omega$$

الأسبوع ٢٦

أب August 8

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Wed.						

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السبت

$$(iii) X_L = 2\pi fL$$

$$L = \frac{X_L}{2\pi f}$$

$$L = \frac{10}{2\pi \times 50} = 0.031 \text{ H}$$

Example (3)

A single phase, 230V, 50 Hz supply down the power 2 h.p. from a.c motor of 12.5 A and Power Factor of 0.7 What is the input power and efficiency

$$V = 230 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$P_{out} = 2 \text{ hp} = 2 \times 746 = 1492 \text{ watt}$$

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الخميس

$$P_{in} = VI \cos \phi$$

$$(i) P.f = \cos \phi$$

$$P_{in} = 230 \times 12.5 \times 0.7 = 2012.5 \text{ watt}$$

$$(ii) \text{ efficiency } \eta = \frac{P_{out}}{P_{in}}$$

$$\eta = \frac{1492}{2012.5} = 0.74136$$

$$\% \eta = 0.74136 \times 100 = 74.136 \%$$

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الجمعة

الأسبوع ٣٥

أب August 8

NOTES

ملاحظات

Sat.	30	23	16	09	02	السبت
Sun.	31	24	17	10	03	الأحد
Mon.		25	18	11	04	الاثنين

27

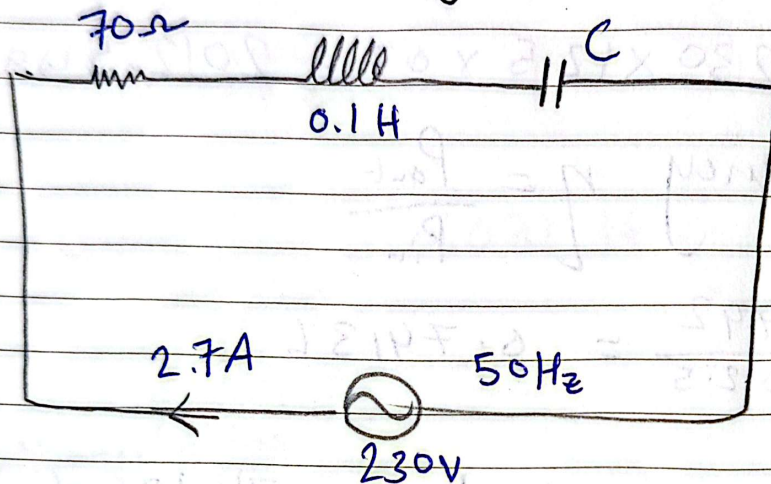
WEDNESDAY

27/8/2025

الأربعاء

Example (4)

Given the following R-L-C Series circuit



Calculate the value of capacitance

$$Z = \frac{V}{I} = \frac{230}{2.7} = 85.185 \Omega$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.1 = 31.415 \Omega$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \text{Squaring both sides}$$

$$Z^2 = R^2 + (X_L - X_C)^2$$

Week 35

NOTES

ملاحظات

أيلول September 9

Sat.	27	20	13	06	السبت
Sun.	28	21	14	07	الأحد
Mon.	29	22	15	08	الاثنين
Tue.	30	23	16	09	الثلاثاء
Wed.		24	17	10	الأربعاء
Thu.		25	18	11	الخميس
Fri.		26	19	12	الجمعة

26

TUESDAY

26/8/2025

الثلاثاء

$$(85.185)^2 = (70)^2 + (31.415 - x_c)^2 \quad 8$$

$$7256.484 = 4900 + (31.415 - x_c)^2 \quad 9$$

$$7256.484 - 4900 = (31.415 - x_c)^2 \quad 10$$

$$2356.484 = (31.415 - x_c)^2 \quad \checkmark \quad 11$$

$$48.54 = 31.415 - x_c \quad 12$$

$$48.54 - 31.415 = -x_c \quad 1$$

$$x_c = 17.125 \quad 2$$

$$x_c = \frac{1}{2\pi f c} = 7 \quad c = \frac{1}{2\pi f x_c} \quad 3$$

$$c = \frac{1}{2\pi \times 50 \times 17.125} = 1.858 \times 10^{-4} \text{ F} \quad 4$$

$$c = 185.8 \times 10^{-6} \text{ F} = 185.8 \mu\text{F} \quad 5$$

الأسبوع ٣٥

أب August 8

Sat.	30	23	16	09	02	السبت
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Tue.		26	19	12	05	الثلاثاء
Wed.		27	20	13	06	الأربعاء
Thu.		28	21	14	07	الخميس
Fri.						

NOTES

ملاحظات

25

MONDAY

25/8/2025

الاثنين

Example (5)

A coil having a resistance of $10\ \Omega$ and inductance of $0.2\ \text{H}$ is connected to a $100\ \text{V}$, $50\ \text{Hz}$ supply calculate!

(a) the impedance

(b) reactance

(c) current

(d) the phase difference between current and voltage

Solution:

$$R = 10\ \Omega$$

$$L = 0.2\ \text{H}$$

$$V = 100\ \text{V}$$

$$f = 50\ \text{Hz}$$

(a) $Z = ?$ (b) $X_L = ?$ (c) $I = ?$ (d) $\theta = ?$

Week 35

NOTES

ملاحظات

أيلول September 9

Sat.	27	20	13	06	السبت
Sun.	28	21	14	07	الأحد
Mon.	29	22	15	08	الاثنين
Tue.	30	23	16	09	الثلاثاء

24

SUNDAY

24/8/2025

الأحد

$$(a) Z = \sqrt{R^2 + X_L^2}$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.2 = 62.831 \Omega$$

$$Z = \sqrt{(10)^2 + (62.831)^2} = 63.621 \Omega$$

$$(b) X_L \approx 62.831$$

$$(c) I = \frac{V}{Z} = \frac{100}{63.821} = 1.511 \text{ A}$$

$$(d) \theta = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\left(\frac{62.831}{10}\right)$$

$$\theta = 80.956^\circ$$

الأسبوع ٣٥

أب August 8

NOTES

ملاحظات

Sat. 30 23 16 09 02 السبت

23

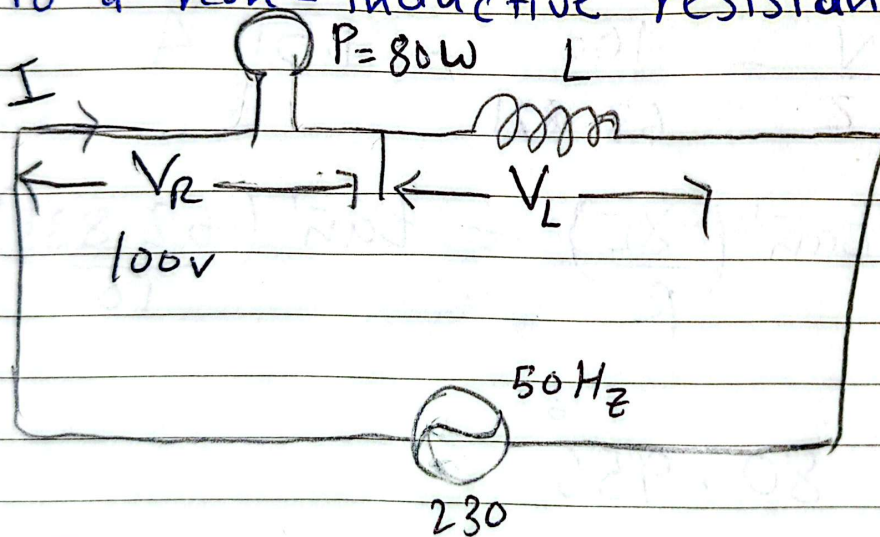
SATURDAY

23/8/2025

السبت

Example (7)

A 100V, 80W lamp is to be operated on 230V, 50Hz supply. Calculate the inductance of the choke required to be connected in series with the lamp for this separation. The lamp can be taken as equivalent to a non-inductive resistance



$$V = \sqrt{V_R^2 + V_L^2}$$

squaring both sides

$$V^2 = V_R^2 + V_L^2 \Rightarrow (230)^2 = (100)^2 + V_L^2$$

Week 35

NOTES

ملاحظات

أيلول September 9

Sat.	27	20	13	06
Sun.	28	21	14	07
Mon	29	--	--	18 01

 السبت
الأحد
الاثنين

21

THURSDAY

21/8/2025

الخميس

$$V_L^2 = (230)^2 - (100)^2 = 4400 \quad \checkmark$$

$$V_L = 207.123 \text{ V}$$

$$P = VI \Rightarrow I = \frac{P}{V} = \frac{80}{100} = 0.8 \text{ A}$$

$$P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} = \frac{(100)^2}{800} = 125 \Omega$$

$$Z = \sqrt{R^2 + X_L^2} \Rightarrow Z^2 = R^2 + X_L^2$$

$$X_L^2 = Z^2 - R^2 \Rightarrow Z = \frac{V}{I} = \frac{230}{0.8}$$

22

FRIDAY

22/8/2025

الجمعة

$$Z = 287.5 \Omega \quad X_L^2 = (287.5)^2 - (125)^2$$

$$X_L = 258.903 \quad L = \frac{X_L}{2\pi f}$$

$$L = \frac{258.903}{2\pi \times 50} = 0.412 \text{ H}$$

الأسبوع ٣٤

أب August 8

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